

High fidelity modeling tools for engine liner design and screening of advanced concepts

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Abstract

With aircraft engines trending toward ultra-high bypass ratios, resulting in lower fan pressure ratios, lower fan RPM, and therefore lower blade pass frequency, the aircraft engine liner design space has been dramatically altered. This result is also due to the associated reduction in both the available acoustic treatment area (axial extent) as well as thickness (liner depth). As a consequence, there is current need for novel acoustic liner technologies that are able to meet multiple physical constraints and simultaneously provide enhanced noise attenuation capabilities. In addition, recent advances in additive manufacturing have enabled the consideration of complex liner backing structures that would traditionally be limited to honeycomb cores. This paper provides an overview of engine liner modeling and a description of the key physical mechanisms, with some emphasis on the use of low to high-fidelity tools such as empirical models and commercially available software such as COMSOL, Actran, and PowerFLOW. It is shown that the higher fidelity tools are a critical enabler for the evaluation and construction of future complex liner structures. A systematic study is conducted to predict the acoustic performance of traditional single degree of freedom liners and comparisons are made to experimental data. The effects of grazing flow and bias flow are briefly addressed. Finally, a more advanced structure, a metamaterial, is modeled and the acoustic performance is discussed.

Keywords

Acoustic liner, acoustic metamaterial, Lattice Boltzmann, liner design, noise reduction

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Introduction

Motivation

Commercial aircraft engines have been trending from low turbofan bypass ratios (mass flow rate of bypass stream to mass flow rate entering the core) in the 1960s to ultra-high bypass ratios 15+ of the future (see Figure 1). Driving this trend is the fact that higher bypass ratio designs can result in a more efficient engine with lower fuel consumption and lower emissions. Noise has been significantly reduced with the higher bypass ratio designs and the dominant noise sources from the turbofan engine have shifted from jet dominated noise to fan noise, as well as the emergence of other engine sources such as the combustor, which traditionally has been less important to low noise designs, but is now trending to be near equal strength and importance (see Figure 2). Ultra-high bypass ratio designs result in lower fan pressure ratios, lower fan RPM, and therefore a lower fan blade passage frequency. Acoustic liners are used throughout the engine as an effective means of reducing the radiated sound and are required to meet aircraft/engine certification goals (see Figure 3). To be most effective, engine liners of the future will need to reduce noise at lower frequencies and over a broader frequency range. Another result of the higher bypass ratio engine designs is that in order to maintain or

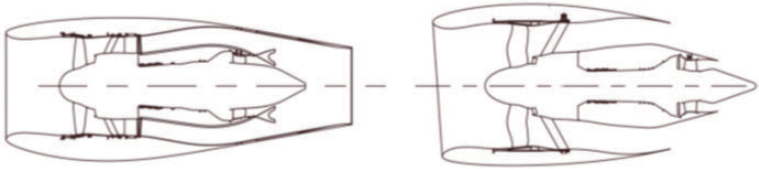


Figure 1. Growth in turbofan engine bypass.¹

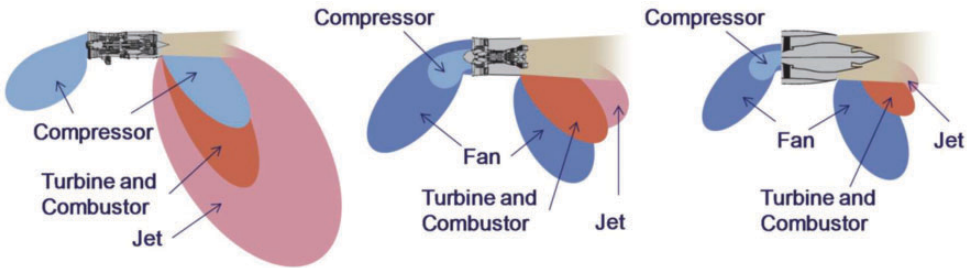


Figure 2. Noise changes with increasing turbofan engine bypass.¹

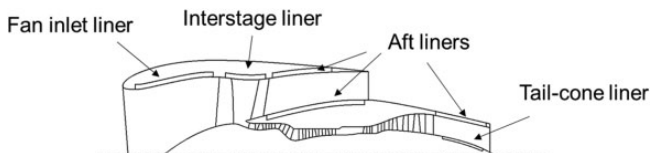


Figure 3. Typical placement of acoustic treatment inside an engine and nacelle.

enhance the fuel burn savings that is negatively impacted by the increased weight and drag of larger fans, the engine nacelle designs need to be more compact, i.e., shorter duct lengths and thinner loft lines. This results in thinner and potentially less axial extent of available acoustic treatment area, thus posing significant challenges to the engine liner design where low frequency and more effective performance traditionally required thicker liners and more available area in the nacelle. Most design approaches have utilized empirically based tools to define the liner construction and geometry and, today, finite element (FE) tools are becoming more commonly used for the propagation and radiation to the far field. Thus, to overcome the noise and fuel burn technical challenges of future aircraft, there is a critical need for the development of next generation acoustic liners along with validated modeling tools to evaluate and to be incorporated within the design processes.

Current industry liner design process

The liner design approach at Collins Aerospace, which is considered representative for the aerospace industry, consists of a three-step process, starting with impedance optimization, then followed by the assessment of the liner attenuation in the far field, and finally the evaluation of system level noise targets (see Figure 4). A multi-variable optimization is performed over a frequency range based on pre-defined attenuation targets and seeks a combination of liner specifications that best match the target. The liner specifications are generally based on honeycomb sandwich panels that are typically used in today's aerospace applications. The target attenuation spectrum is defined, prior to the optimization, in

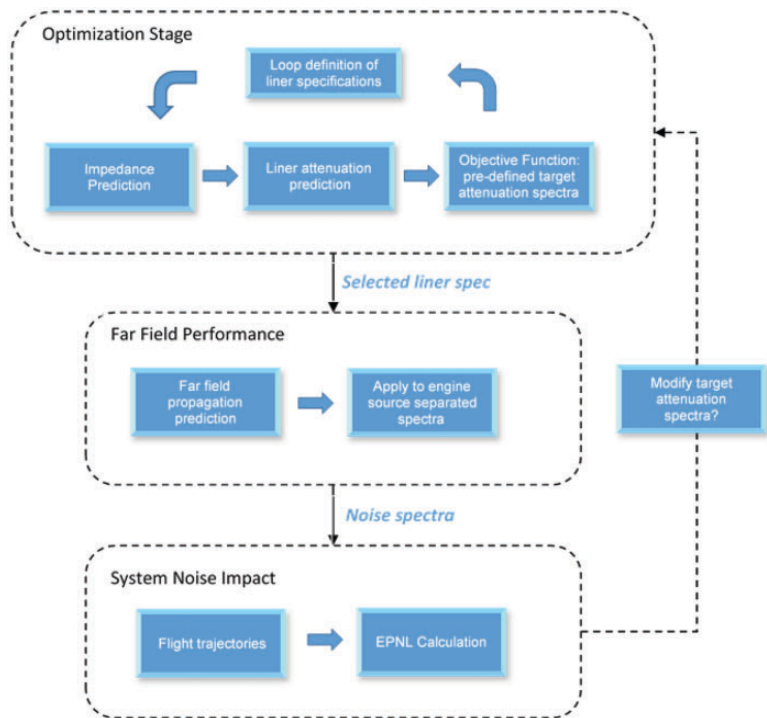


Figure 4. Summary of the liner design process.

concurrence with the engine manufacturer's system level noise targets. Once an optimum liner is found, the corresponding impedance is utilized to evaluate the liner performance on the far field noise, which allows re-adjusting the engine manufacturer's proprietary noise database and performing effective perceived noise level (EPNL) estimations. This last step is normally conducted in collaboration with the engine manufacturer.

The commercially available FE acoustic prediction software Actran has been incorporated into this liner optimization processes. The solution methodology has proven quite reliable in estimating noise propagation and attenuation in complex lined ducts in the presence of realistic grazing flows and offers approximations that favor computational time while still providing accurate results. An Actran model generally requires the creation of a computational mesh, a realistic CFD flow database, and the assembly of a finite element input file to specify the noise excitation and boundary conditions. For most aerospace industry applications, the liner impedance, as a function of frequency, must also be available from a reliable methodology, such as grazing flow impedance tests or accurate validated prediction models. A representative reduced order model (ROM) for predicting the liner impedance is the Goodrich impedance model that is currently used by the Collins Aerospace nacelle business to perform liner optimizations along with pre-test and pre-certification noise predictions in support of the engine manufacturer's system level noise estimations. The Goodrich model is based on a modified Crandall's solution² and empirically adjusted to more accurately quantify the discharge coefficients of realistic perforate hole profiles, such as tapering effects, obtained with different manufacturing methods. The model is well-documented in the open literature.²⁻⁶ As described by Kraft et al.,³ the successful implementation of this model depends on experimentally acquired values for effective percent open area (POA) and hole diameter obtained from DC flow resistance data. These parameters are typically used for industry material qualification. The model is also valid for linear (wire mesh) liner applications. In addition, the model has been used in conjunction with test data to conduct effective POA studies to support quality control, novel liner developments, and, in certain occasions, perform no acoustical change certification analysis. The normalized impedance $z = \theta + i\chi$ (where θ is the normalized resistance and χ is the normalized reactance) of this model consists of three main contributions: (1) a linear term (index "*l*"), associated with the viscous flow through the orifices, (2) a non-linear term (index "*nl*"), associated with high amplitude acoustic waves leading to the formation of a jet flow through the orifices with vortex ring generation and turbulence breakdown, and (3) a linear term accounting for the effect of the grazing flow over the facesheet (index "*gf*"). Through the use of this index nomenclature, the structure of the overall model can be written as follows:

$$z = (\theta_l + \theta_{nl} + \theta_{gf}) + i(\chi_l + \chi_{nl} + \chi_{gf}) \quad (1)$$

Details of the different terms will be discussed in later sections, in connection with the comparisons to the high-fidelity simulations.

The modeling of traditional honeycomb design acoustic liners and its application in flow ducts, grazing flow facilities, and turbofan engines continues to be improved and validated. The following is an example of a typical prediction conducted at Collins Aerospace, in combination with the Goodrich impedance model, to study different liner configurations in its 4 in by 5 in rectangular insertion loss flow duct facility. The presented case corresponds to a single degree of freedom (SDOF) honeycomb liner tuned in the frequency range of

1500 Hz to 2000 Hz with facesheet porosity targeting broadband attenuation capabilities. The results are presented as the predicted insertion loss compared to the flow duct measurements performed with broadband noise excitation and two mean flow scenarios: no flow and Mach number 0.5, as shown in Figure 5. The input source for the predictions utilizes equal energy modal distribution. As seen in the comparison, the prediction captures well the main attenuation peak due to the first liner resonance as well as the overall broadband attenuation trend over the entire frequency range. This methodology is useful for comparative investigations of the frequency response of novel concepts relative to the more traditional honeycomb liners.

Goals of the paper

The liner design process outlined above and the example application rely on suitable impedance models to represent the liner structure in an efficient and simple way. The Goodrich ROM (equation (1)) has been well-established for the traditional, more conventional, SDOF liners. As noted above, the future technology trends require more complex liner structures, where established ROMs for SDOF and more traditional liners become less applicable, thus making it necessary to establish models and simulation approaches that are capable of serving the same purpose for future, more complex, liner technologies. Once such approaches are established, the design process outlined above can be used in conjunction with an updated impedance model. The focus of this paper is on assessing the suitability of advanced, higher fidelity simulation approaches for liner impedance predictions, relevant to advanced liner concepts. In particular, the objectives are to (1) document the use and validity of current tools along with identifying the fundamental physics, (2) in a systematic way, establish tools/approach for efficient and effective designs and evaluations of future novel and more complex liners, and (3) demonstrate, with a simplified example, the performance of one such novel liner concept.

Literature review

Review intent

This section is to provide a review of the state of liner acoustic modeling, along with identifying key insights obtained from this body of work to guide the modeling of advanced liner

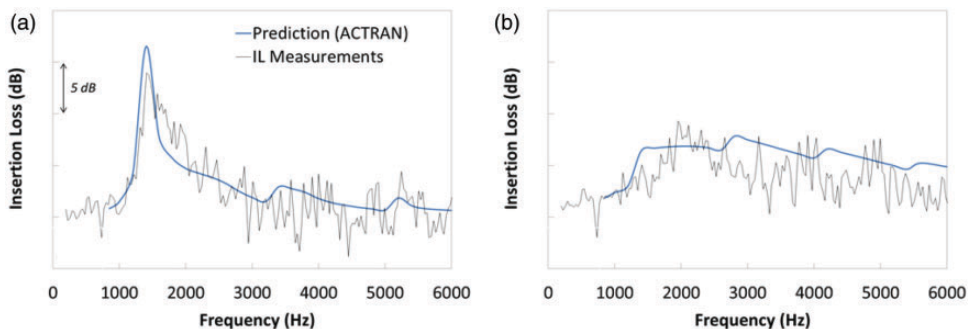


Figure 5. Comparison of insertion loss measurements and predictions for a traditional SDOF liner at (a) no mean flow, (b) $M = 0.5$.

concepts. Two broad categories will be distinguished here: mid-fidelity and high-fidelity simulation approaches. The former relies on simplifying assumptions like acoustic wave equation solvers with embedded sub-models for perforations, or linearized Navier-Stokes equations, for example. For these approaches, commercially available software packages, such as COMSOL or Actran, are commonly used. High-fidelity simulations, in comparison, are based on a direct resolution of the liner perforations and geometry details in the numerical grid and can involve the solution of unsteady RANS equations, or more commonly Large Eddy Simulation (LES) or Direct Numerical Simulation (DNS), to capture the local unsteady flow effects inside the perforation holes under acoustic excitation.

Many of the studies referenced in this section utilize a SDOF liner design as a baseline for comparison and validation. There is a significant amount of data available in the open literature pertaining to the acoustic performance of such liners. Key components of SDOF liners include the perforate facesheet, honeycomb core, and backing plate (assumed rigid), see Figure 6. The perforate facesheet geometry is characterized by its percent open area (POA or σ), hole diameter (d), and facesheet thickness (t). The honeycomb core depth (D) is an important parameter, as it is a key factor in setting the resonance response of the liner material, and the honeycomb cell size (hexagonal area) is sized such that the liner can be considered locally reacting. Multiple degree of freedom (MDOF) liners are similarly constructed, but include a septum layer or layers that divide the individual honeycomb core cavities.

The geometric properties of an acoustic liner can be varied based on the application and as noted above, over the years researchers have made efforts to develop simulation tools and models to represent the underlying physics associated with those geometric features. The main findings from such studies are reviewed next.

Mid-fidelity approaches

Schiller et al.⁷ evaluated COMSOL Multiphysics, a commercially available finite element software, for modeling acoustic liners in a grazing flow setup. The authors studied so called “reaction liners” which incorporate back cavity volume without lateral partitions as opposed to traditional liners with honeycomb cells. The convected Helmholtz equation was solved in the fluid domain, with the mean flow calculated in a prior step using a

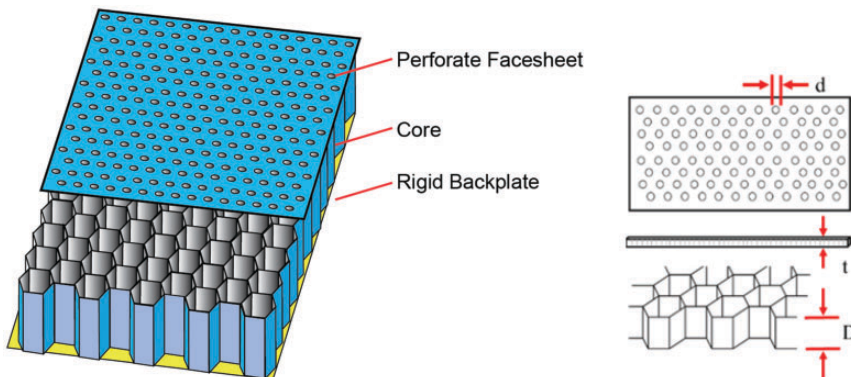


Figure 6. SDOF liner with perforate facesheet, honeycomb core, and backing plate.

compressible potential flow solver. The perforate facesheet was modeled as a transfer impedance within the fluid domain, with empirically determined parameters. Fibrous bulk material inside the cavity, intended for broadband attenuation, was modeled as an equivalent fluid with complex density and complex speed of sound which were determined experimentally. The simulation results correlated well with the corresponding experimental data obtained by NASA Grazing Flow Impedance Tube (GFIT) test measurements. In an effort to increase the computational fidelity for liner impedance calculations using COMSOL Multiphysics, Jensen et al.⁸ followed a similar two-step process, where first a RANS simulation with SST turbulence model was performed for the background steady grazing flow. A frequency domain version of the linearized Navier-Stokes equations was then solved in a second step to include viscous and thermal effects. The simulation process was demonstrated on a 2D model of an acoustic liner represented by an array of eight SDOF resonators in a duct with grazing flow. The axial distribution of the sound pressure above the liner was reported but no direct comparisons to experimental data were made. In a subsequent study, Jensen et al.⁹ incorporated an impedance eduction method for this setup. The eduction process was verified, but comparison to experimental data showed a rather large overprediction in both resistance and reactance, possibly related to the way the impedance boundary condition was implemented.

Sugimoto et al.,¹⁰ utilized the acoustic wave equation finite element solver Actran to design and evaluate the acoustic impedance of a 2D folding cavity liner concept compared to traditional SDOF and double-degree-of-freedom (DDOF) liners. The interfaces between the liner segments with perforations were modeled using a transfer impedance. The resistance and mass reactance of the facesheet and septum were specified that way. Predicted impedance of the folding concept compared well to a SDOF liner with depths associated with the back wall before the bend and one with an effective length that includes the fold. Grazing flow results showed good agreement between this folding liner concept and the DDOF liner demonstrating the effectiveness of the folding concept over a wide frequency range. In a similar effort, Marinova et al.¹¹ reported on a code developed to assist in the design of variable-depth, bent-chamber acoustic liner that targets broader frequency acoustic attenuation than conventional single-depth chamber liners. The proposed liner design targets broadband frequency attenuation by using multiple resonator chambers with dissimilar lengths which provide peak absorptions at multiple frequencies, thereby broadening the attenuation frequency range. Sizing and fitting these multiple variable-length chambers, which may be also bent or folded within a given volume, is a challenge. The developed code automatically determined the most efficient way of “packing” a given volume with variable-length chambers. The utility of this code was demonstrated by designing and fabricating an acoustic absorber. These measured absorptions were matched reasonably well by model predictions which were obtained both by a transmission type (1D) model as well as a 3D COMSOL model.

High-fidelity LES approach

Mann et al.¹² performed 3D numerical impedance tube liner calculations using the Lattice Boltzmann Method (LBM) solver PowerFLOW that is based on a very large eddy simulation (VLES) turbulence concept. A source excitation of 120 dB overall sound pressure level (OASPL) was introduced upstream of the liner sample as an initial condition. The reflected wave signal was then analyzed at several upstream microphone locations to calculate the

impedance. A grid resolution study showed that for a SDOF liner at least 15 voxels (i.e. volume elements or computational cells) per orifice diameter should be used to obtain a grid-independent solution. A systematic study was then performed, investigating the liner absorption behavior for variations in facesheet thickness, hole diameter, and backing cavity depth. The numerical results were compared to two semi-empirical models with largely consistent results, but no comparison to experimental data was made. Additional simulations were performed for a DDOF liner and for a bulk absorber. The latter considered a model to account for the bulk porous medium via a momentum sink term based on Darcy's law. Comparisons between this reduced-order LBM porous medium simulation and a semi-empirical model showed very good agreement. Overall, this study has shown that high-fidelity normal impedance liner simulations, including SDOF, DDOF, and complex structures can be evaluated; although, experimental validation was recommended by the authors. Hazir and Casalino¹³ used the established LBM impedance tube simulation approach to study DDOF liners with a heated or cooled septum, as a novel approach to make the liner adaptive for different operating conditions and more compact. It was demonstrated by simulations that with a cooled septum, -100 K temperature difference, the depth of the secondary cavity in the DDOF liner could be reduced by 26%. For a heated septum, $+100$ K temperature difference, the liner width could be increased to bring the increased frequency back to the original target value. In this case, 17.5% fewer honeycomb cavities would be needed for the original design target frequency. No additional validation with experimental data was conducted for this exploratory study.

Based on the established LBM simulation approach for zero-flow impedance tube, Casalino et al.¹⁴ simulated a full 360 degree annular section inlet liner inside a turbofan inlet region under grazing flow, by resolving each individual liner hole. The liner was analytically optimized for maximum absorption at 1BPF and 2BPF of the fan tonal noise. This demonstration was intended as a conceptual exploratory study; a grazing flow liner performance prediction study had not been performed at that point in time. The authors demonstrated for the first time the capability to include a SDOF liner in its full geometrical complexity in a LBM-based VLES turbofan simulation.

Manjunath et al.¹⁵ extended the study by Mann et al.¹² towards a grazing flow facility environment, but still without grazing mean flow. The NASA GFIT setup was mimicked in the LBM simulation. A single cell of a SDOF liner was simulated with either a single facesheet hole (0.99% porosity) or 7 facesheet holes (6.9% porosity) to study the impact of hole interactions. Grazing acoustic waves over the liner were simulated. The impedance was calculated in-situ using Dean's method¹⁶ based on two virtual microphones. A grid refinement study showed that ~ 42 voxels per orifice hole diameter were sufficient to get grid-independent impedance values. This number was found to be much larger than the value of 15 determined by Mann et al.¹² for the normal impedance tube setup. It was pointed out that both resolution per orifice diameter and per facesheet thickness are important. Comparison of the LBM-predicted single orifice (0.99% porosity) resistance and reactance results with experiments and the same semi-empirical models used by Mann et al.,¹² showed overall good agreement among all three. The effect of SPL was studied as well and ranged from 120 dB up to 160 dB. The predicted values compared well with experimental data. The occurrence of vortex rings with associated turbulent breakup for an SPL of 140 dB was found to be a driving mechanism for the increased resistance, which is in accordance with the literature. Avallone et al.¹⁷ further extended the work by Manjunath et al.¹⁵ by including a grazing flow at $M_{gr}=0.3$. Simulations for the same 6.9% porosity liner were performed at

130 dB source amplitude for a single excitation frequency. A turbulent boundary layer trip was placed far upstream of the single cell liner in the simulation. The trip was used to switch the LBM VLES turbulence model from URANS to scale-resolving mode. The finest resolution simulations used 30 voxels per hole diameter, as a compromise for acceptable computational cost and accuracy. Resistance and reactance (averaged over all orifice holes) were found to agree well with experimental data. The orifice hole-by-hole impedance calculation showed significant variations in the predicted resistance, indicating the importance of flow interaction between the downstream holes. A deeper investigation into the flow physics of ejecting vorticity was performed by the authors to understand flow interaction mechanisms.

Dassé et al.¹⁸ compared their in-house LES code, AVBP, to existing analytical models and experimental data. As part of their work the authors looked at perforated plates for the application to combustion noise. In particular, their work focused on understanding acoustic absorption and the impact of flow through the perforated sheet (i.e. bias flow). The modeling effort and model comparisons were for a simplified geometry, which consisted of a single perforation hole and back cavity. The authors showed that the presence of an acoustic source at different frequencies had an impact on the absorption levels and that differences were observed when the perforated plate thickness was considered, especially at higher frequencies. The authors noted that high-fidelity LES modeling can be used to explore the impact of plate thickness and more complex features such as hole angle, in the context of vortex (i.e. flow) and acoustic interaction.

The work of Ji and Zhao¹⁹ was motivated by thermoacoustic instabilities in a combustor, where damping from perforated liners is used to mitigate the coupling of heat release and acoustic waves in the combustion chamber. A 3D LBM simulation approach was used to observe the time evolution of vortex shedding at a single orifice and to characterize the corresponding acoustic damping. The effect of bias flow, mean flow through the orifice, was studied to simulate the needed air cooling in a real combustor liner. Their model compared very well with experimental work from the authors including the model resistance and reactance dependence on both amplitude and frequency of the incident sound. For single frequency excitation, it was shown that an unsteady round jet is generated with vortex shedding at the excitation frequency and the subsequent conversion to kinetic energy that provides effective sound absorption. For multiple tone excitation, the absorption compared very well to several existing orifice models, including Howe's model,²⁰ below the peak absorption. Above the peak absorption it was noted that plate thickness contributed to the differences observed and concluded that the thickness is indeed an important parameter for predicting the orifice damping performance.

High-fidelity DNS approach

Tam et al.²¹ investigated the concept of vortex shedding as the dominant dissipation mechanism of resonant acoustic liners under certain conditions. It was pointed out that Melling³ proposed that the main dissipation mechanism for low sound intensity forcing was wall friction around the resonator opening and when the sound intensity became high, an oscillatory turbulent jet would develop at the opening. The authors observed that vortex shedding at the mouth of the resonator occurred at moderate to high incident sound intensity, associated with a significant increase in energy dissipation. The authors performed both 2D numerical simulations and normal incidence impedance tube experiments of a slit resonator. Overall good agreement between the numerical simulations and the experiments was found.

The relative importance of grazing flow versus sound intensity was not considered for this study in the context of the vortex shedding dissipation mechanism described. Tam et al.²² later extended the study towards a 3D simulation for SPL ranging from 114 dB to 145 dB. The observed vortex shedding was different in structure to the 2D simulations but was still found to be the dominant mechanism at high SPL. Good agreement with normal impedance tube data was found. In a subsequent study, Tam et al.²³ simulated grazing flow at $M_{gr}=0.3$ over the 2D slit-resonator liner with overall good agreement to experimental data, albeit some larger differences in predicted resistance at low frequencies. The authors also identified liner self-noise production due to acoustic feedback mechanisms over the mouth of the resonator slits, but this occurs at sufficiently high frequencies above the hearing range. Implications of the observed flow field on drag were discussed as well. While 2D simulations are helpful in understanding some of the key physics associated with the liner absorption behavior, the results cannot always be translated to the realistic hole geometries for liners used in practice.

A comprehensive numerical study of an acoustic liner with a single circular orifice and no flow was conducted by Zhang and Bodony.²⁴ The liner was subjected to incident acoustic waves with amplitudes of 130 dB to 160 dB. The authors discussed how the acoustic forcing amplitude and frequency plays a role in the amount of flow discharged, with vortical structures being formed, out of and into the orifice/liner. The authors highlighted the importance of correctly capturing the in-orifice velocity profiles for an accurate time-impedance model. The authors subsequently extended their study to investigate normal-incidence acoustic wave interaction with the single-orifice SDOF liner in presence of grazing flow.²⁵ Detailed DNS studies were performed for both laminar and turbulent grazing flow over the liner orifice. The incident SPL was again varied from 130 dB to 160 dB. A detailed analysis was performed to understand the unsteady flow through the orifice and the relevant velocity scales involved and the implications on liner drag. Based on the DNS database that was created, the authors included the grazing flow effect in a reduced-order time impedance model.

Review summary

A select sample of papers were reviewed and summarized pertaining to a set of tools and approaches, at varying fidelity, to define the physics as well as to capture some of the learning and concepts that researchers have obtained and proposed for the design and analysis of turbofan engine acoustic liners. Reduced order empirical models have evolved over the decades and have proven to be quite adequate for supporting the more traditional conventional engine liner evaluations and overall design process. Industry has adopted and or developed these types of models to suit their needs and products. However, as the newer liner designs have trended away from the more traditional facesheet-honeycomb designs, due to the trends in the aircraft engine architectures, the tools used have to include more fidelity. The use of mid-fidelity tools such as Actran and COMSOL have provided researchers the ability to explore in more detail the physics of not only traditional, but also the more sophisticated liner designs. These tools are capable of either using an impedance boundary condition to represent the liner or simplified sub-model to capture the leading order acoustic effects of facesheet and septum perforations. These tools/models are obviously more computational expensive, as compared to the ROMs, but they expand the liner design space. Comparisons to experimental data have shown that these tools can be used with confidence,

but when considering the presence of high amplitude acoustic forcing and unsteady flows, grazing or bias, there is the need for the higher-fidelity tools/methods. The higher-fidelity methods also allow researchers to explore reduced domains, from say the liner coupon size to a smaller portion of the domain where only one or a small number of facesheet holes and a single cavity is included. This enables for a deeper more focused exploration of the physics of both the traditional and more advanced liner designs, especially for topics like the acoustic wave and unsteady flow interaction at the facesheet hole and the transfer of energy from acoustic to vortical motion. More efficient methods are being developed to reduce the computational cost of high-fidelity methods, but they are still not at the point that they can be used during the engine liner design process. However, as covered in the high-fidelity approach sections above, the LBM VLES can be considered a good compromise between fidelity and computational affordability. DNS studies remain the gold standard for very detailed and simple studies but will not in the foreseeable future be able to address complex liner geometries. The LBM VLES simulation results to-date are very promising, but the number of studies is still very limited as the method is relatively new. A comprehensive study to establish this approach for the different acoustic absorption mechanisms at high SPL and in presence of flow, with possible interaction effects, has not been undertaken yet and is needed to build further confidence in the predictive capabilities.

High-fidelity liner simulations

Introductory remarks

In this section the suitability of high-fidelity simulations based on the LBM VLES approach is evaluated to determine if this approach is able to adequately predict the different acoustic absorption mechanisms of a perforate acoustic liner. LBM simulations are performed for the NASA Langley Research Center's (LaRC) Normal Incidence Tube (NIT) and Grazing Flow Impedance Tube (GFIT) setups and the results are compared to available NASA experimental data and to ROMs, such as the Goodrich model, equation (1), mentioned above. The intent of this study is to build confidence in and demonstrate the numerical tools that can be used for advanced liner analysis by first evaluating the normal incidence results (resistance, reactance, and absorption) followed by grazing and bias flow studies. For the purpose of this paper, the presented simulation cases are representative but non-exhaustive. The focus is placed on the overall prediction of resistance, reactance, and acoustic absorption to enable the evaluation and trade studies for industry applications, rather than rigorously interrogating the high-fidelity simulations for new physical insight into the key absorption mechanisms, which is something to be reserved for future studies.

Experimental data sets

Impedance data (resistance, reactance, and absorption) from the NASA LaRC NIT and GFIT are used for comparison with the modeling results to follow (Figure 7). This data, primarily for the liner configuration with a 6.4% POA, is taken from the work of Jones et al.,^{26,27} where the results are conveniently tabulated for both the no flow and with grazing flow conditions. Detailed facility descriptions and all test conditions are also provided in the literature.

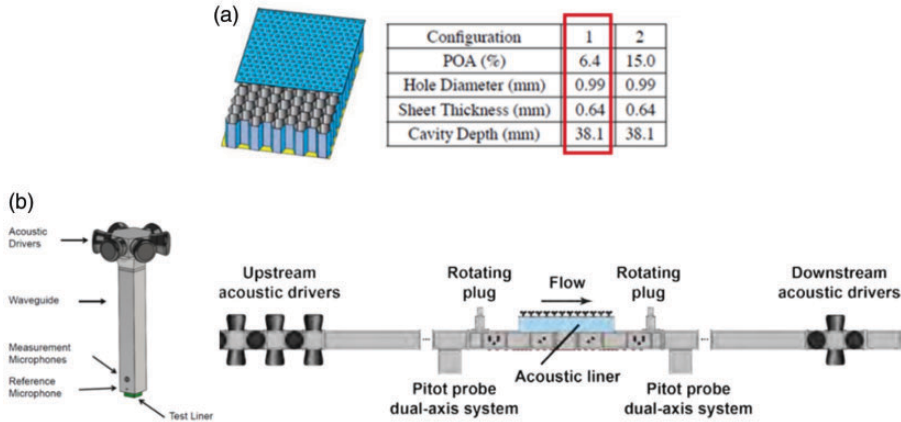


Figure 7. SDOF liner and facilities used for comparison. (a) SDOF liner properties²⁶ and (b) NASA NIT and GFIT schematics²⁷.

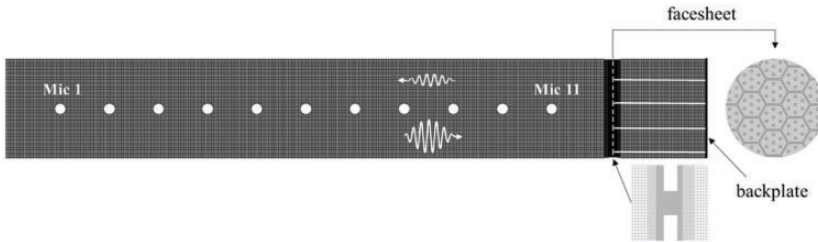


Figure 8. 2D cut-plane view of the 3D numerical impedance tube SDOF liner simulation setup.

No flow

A 3D numerical impedance tube setup as shown in Figure 8 is considered for impedance prediction of the SDOF Configuration 1 liner from Figure 7(a), with no flow. The simulations are performed with the LBM code PowerFLOW V6-2019, which is capable of predicting the acoustic wave absorption mechanism by resolving the actual facesheet geometry in the numerical grid. Based on best practices from previous simulations using this approach,^{12,15,17} a grid resolution of 30 voxels (i.e., volume elements) per perforation hole diameter was chosen. A cylindrical cut-out of a larger liner geometry is simulated with no-slip hard wall boundary conditions, as shown in Figure 8. The duct is sized such that only plane wave modes are cut on. Away from the orifices the grid is coarsened and sized to capture the turbulent flow leaving the orifices; the grid remains comparatively fine for studying acoustic propagation (~ 134 voxels/wavelength for the highest frequency considered here (5 kHz)). The overall numerical grid consists of 74.8 million voxels. The simulated cylindrical duct is closed off at all ends with hard walls and the acoustic excitation is introduced upstream of the liner facesheet via an initial condition.

For the first simulation case, the linear behavior of the liner is investigated at a sufficiently low SPL where the solution is expected to be independent of the excitation level.

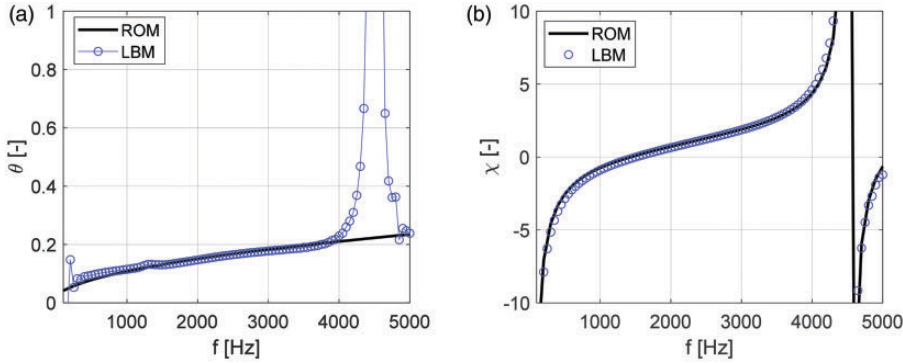


Figure 9. SDOF liner impedance predictions from high-fidelity LBM simulations in comparison to an analytical linear impedance model (equation (3)). Simulations were performed with a broadband source excitation with OASPL = 130 dB.

Based on the available experimental data, acoustic waves consisting of a broadband source with flat excitation amplitude from 50 Hz to 5 kHz and OASPL of 130 dB are injected upstream of the SDOF liner. The source propagates downstream to interact with the liner facesheet first, gets partially transmitted through it and gets reflected at the backplate of the cavity leading to subsequent interactions between facesheet and cavity backplate until all energy content is either reflected or absorbed by the SDOF liner facesheet. The acoustic impedance is calculated at eleven microphone locations upstream of the liner facesheet, using the single microphone method, as shown below:

$$z_j = \theta_j + i\chi_j = \frac{1 + R_j}{1 - R_j} \quad \text{with} \quad R_j = \frac{\hat{p}_j - \rho c \hat{u}_j}{\hat{p}_j + \rho c \hat{u}_j} e^{-2ik\Delta x_j} \quad (2)$$

where $\hat{p}_j$ and $\hat{u}_j$ are the Fourier-transformed pressure and acoustic particle velocity at microphone location j , and Δx_j is the distance between microphone j and the facesheet. $k = \omega/c$ is the acoustic wavenumber. This approach has been established in previous efforts by Mann et al.¹² The results showed that there was little influence due to the Δx_j values used. This was confirmed in the present study, but results were nevertheless averaged over all microphone locations. Simulation results are shown in Figure 9 and compared to a linear impedance model, which is based on Crandall's solution for viscous flow through a narrow tube, with end corrections for flow scrubbing over the facesheet surface.²⁸ This model corresponds closely to the linear term used in the Goodrich model:

$$z_l = \frac{8\mu t}{\rho c \sigma a^2} \left(\sqrt{1 + \frac{(k_s a)^2}{32}} + \frac{\sqrt{2} k_s a^2}{2t} \right) + i \left\{ \frac{\omega t}{c \sigma} \left[1 + \left(9 + \frac{(k_s a)^2}{2} \right)^{-0.5} + 2 \frac{8a}{3\pi t} \right] - \cot(kD) \right\} \quad (3)$$

where $k_s = \sqrt{\omega \rho / \mu}$ is the shear wavenumber. It can be seen in Figure 9 that the impedance predictions agree very well over a large frequency range, with the exception of the resistance

at 4500 Hz, where the second zero-crossing of the reactance occurs, associated with anti-resonance ($kD = \pi$). Here, the facesheet behaves like a hard wall and the resistance has a finite, but very large value. This effect on resistance is not captured in the ROM, but it is also of little relevance for traditional liner designs, as this behavior occurs only near anti-resonance. Except at anti-resonance, the simple liner impedance model appears to be accurate for sharp-edged orifice facesheets at low SPL, and the high-fidelity simulations are capable of predicting the correct liner behavior. Available broadband excitation NASA data²⁷ was not available for 130 dB, but for 120 dB OASPL; it shows similar resistance and reactance values and trends (albeit slightly lower resistance, as expected), but the data only goes out to 3 kHz. More detailed effects, such as interactions between neighboring orifices and different orifice edge geometries can be evaluated this way, but are not considered here.

For the second simulation case, the source type was changed to a single frequency truncated sine wave source with a SPL of 130 dB, instead of the broadband source, to match the experimental conditions of Jones et al.²⁶ as best as possible. The high-fidelity simulations are compared against the NASA experimental data and against the Goodrich model which includes both linear and non-linear contributions. The linear term corresponds closely to the formulation shown in equation (3). The non-linear impedance term is an empirically updated version of Melling's theoretical model³:

$$z_{nl} = \left[\frac{1.336541(1 - \sigma^2)}{2c} \frac{1}{\sigma^2 C_d^2} - i \left(2.07e-2[s] \frac{k}{\sigma^2} \right) \right] u_{rms} \text{ with } u_{rms} = \sqrt{\sum_f \hat{u}^2} \text{ and } |\hat{u}| = \frac{|p_{ref} 10^{\frac{SPL}{20}}|}{|z| \rho c} \quad (4)$$

where u_{rms} is the root-mean-square value of the acoustic particle velocity in the orifice, and C_d is the discharge coefficient, which has the following empirical form:

$$C_d = 0.80695 \sqrt{\sigma^{0.1} e^{0.5072t/d}} \quad (5)$$

This model requires an iterative solution, due to the dependency of z_{nl} on the acoustic particle velocity $\hat{u}$. Figure 10 shows the prediction for the selected frequencies that were simulated with the high-fidelity LBM approach. Very good agreement is found between all methods for the reactance. The LBM resistance predictions also agree very well with the NASA²⁶ data; although, the ROM predictions from equation (4) show a much more pronounced peak near resonance (~ 1500 Hz). This peak is neither present in the NASA²⁶ data nor in the LBM simulations, which both show a flatter response. It is interesting to note that the NASA results reported in the 2020 TP²⁷ do show this “hump” in resistance response at resonance. That data was for 120 dB and 140 dB for tones and broadband and possibly a different perforated liner. From the ROM, equation (4), it is clear that the source type (broadband or tonal) influences the results because u_{rms} depends on the particle velocity at all frequencies present in the acoustic excitation. Consistent with the NASA results, the source type (tone or broadband) does matter and should be handled based on the application.

The high SPL effect is investigated by comparing LBM simulations with broadband source of different OASPL values. The results are provided in Figure 11 and show an overall

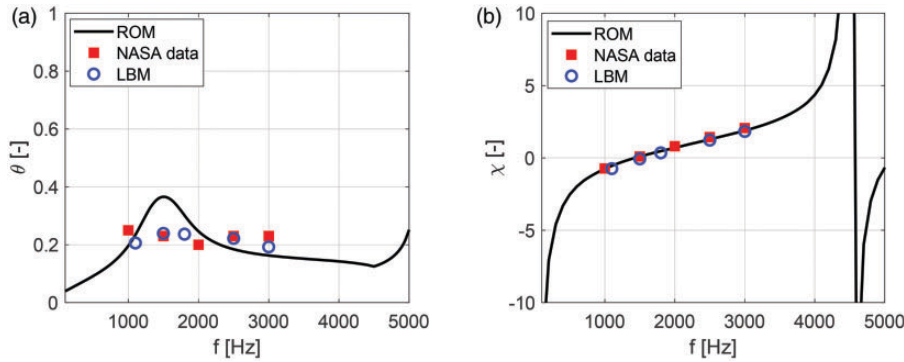


Figure 10. SDOF liner impedance predictions for truncated sine waves with SPL = 130 dB using high-fidelity LBM simulations, compared to a semi-empirical non-linear ROM (equation (4)) and NASA NIT experimental data.

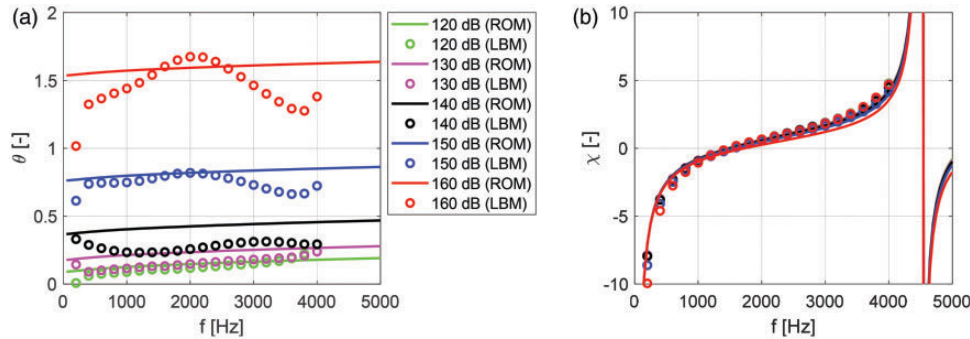


Figure 11. High SPL SDOF liner impedance predictions using high-fidelity LBM simulations with broadband source of different OASPL, in comparison to the ROM (equations (3) and (4)).

reasonable agreement for the predicted resistance for the two higher OASPL values of 150 dB and 160 dB, where the LBM results show a pronounced hump at 160 dB that is absent in the ROM predictions. There is a larger mismatch between the models for 130 dB 140 dB, while good agreement is observed again for 120 dB. It appears that the ROM predicts an earlier onset of the non-linear regime, showing already a noticeable difference between 120 dB and 130 dB results, where the LBM results indicate little difference. Indeed, the good level of agreement between LBM and linear ROM at 130 dB (recall Figure 9) implies that the non-linear regime is more likely to start above 130 dB. For practical purposes, the observed difference has little relevance as nacelle liners are typically designed for much higher resistance values with the presence of grazing flow. The predictions for the reactance are all in very good agreement but indicate a slight flattening of the reactance with increased SPL. For the purpose of this paper one may conclude that the overall level of agreement between high-fidelity LBM simulations and the ROM is satisfactory.

Grazing flow

The high-fidelity simulations are now extended to include the effect of grazing flow. The corresponding simulation setup mimics the NASA GFIT facility. The following process and procedure is based on the work by Avallone et al.¹⁷ The simulation domain consists of a long rectangular duct with spanwise periodic boundary conditions. A single SDOF liner element (Configuration 1 from Figure 7(a)) is placed inside the domain and is exposed to the grazing flow (Figure 12). A velocity boundary condition is set at the inlet to control the freestream Mach number of the grazing flow, and a static pressure condition is used at the outlet of the domain. Upstream and downstream of the domain, numerical sponge zones are used with increased viscosity to damp out acoustic waves to avoid reflections from the domain boundaries. The wall opposite to the liner is treated as a slip wall. Two flow scenarios are initially considered in order to determine the importance of the turbulent flow modeling: (1) A wall-bounded fully developed turbulent boundary layer, where the VLES turbulence model remains in scale-modeling mode and the flow field results closely resemble those of a $k-\varepsilon$ steady RANS solution,²⁹ and (2) a boundary layer trip is specified far upstream in the domain (see Figure 12). For scenario 2, the trip is placed 1755 mm upstream of the liner and has a thickness of 1 mm. The trip height is resolved with four voxels, which was found to be sufficient to trigger and sustain the scale-resolving mode of the VLES turbulence model in the spatially developing turbulent flow downstream of the trip. In both cases the upstream turbulent flow is resolved with $y^+ \sim 28$. Upon reaching the liner, the grid is refined, and 31 voxels are used across each of the seven orifices in the liner facesheet. The overall computational grid consists of 117.7 million voxels. In both scenarios, the simulation is run sufficiently long for the flow field over the liner to establish. A flow solution checkpoint file is written and a single frequency truncated acoustic wave consisting of several wave cycles is imparted on the flow as an initial condition for a subsequent turbulent flow simulation with a superimposed downstream traveling acoustic wave. This finite wave signal is recorded on the facesheet and backplate of the SDOF liner as it passes over it. The acoustic impedance is then calculated using the two-microphone method of Dean¹⁶:

$$z = \frac{\hat{p}_{fs}}{\rho c \hat{u}_{fs}} = \frac{\hat{p}_{fs}}{\hat{p}_{bp}} \frac{-ie^{i\varphi}}{\sin(kD)} \quad (6)$$

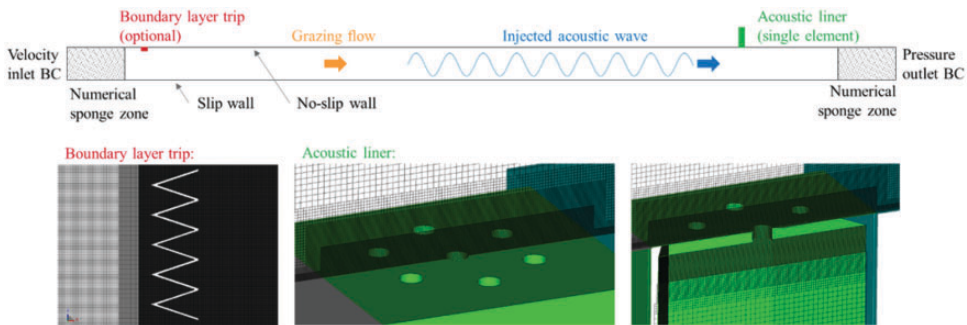


Figure 12. LBM grazing flow simulation setup and computational grid.

where φ is the phase angle between the facesheet pressure $\hat{p}_{fs}$ and the backplate pressure $\hat{p}_{bp}$. The SDOF liner has seven orifices per unit cell (see Figure 13) and the pressure and phase data have been averaged over all holes.^{15,17}

The difference between both simulation approaches is evident by comparing the instantaneous flow fields, as shown in Figure 14(a) and Figure 15(a). Without the boundary layer trip, the flow remains steady and no turbulent fluctuations are resolved. The corresponding time-averaged boundary layer profile is that of a fully developed turbulent flow and matches the NASA experimental data well. The introduction of the boundary layer trip switches the turbulence model into scale-resolving mode (Figure 15(a) and (c)). The time-averaged boundary layer profile is slightly different and does not match the NASA data in the near-wall region as well. However, it is assumed here that the defining parameter to describe the grazing flow boundary layer effect on acoustic impedance is the displacement thickness, which remains largely unaltered between the two different types of flow simulations. This assumption is consistent with observations made by Gabard³⁰ and the empirical ROM that will be applied here and given by equation (7). The difference in boundary layer profile observed with the scale-resolving simulation is therefore expected to have little impact on the predicted acoustic impedance.

To understand the level of unsteadiness introduced by the grazing turbulent flow of the liner in the presence of acoustic forcing, a pressure monitor probe on the facesheet and one

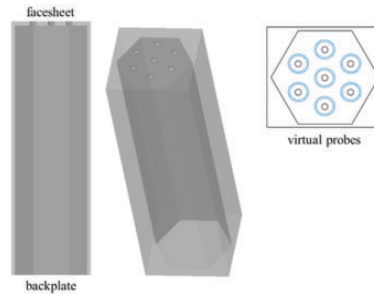


Figure 13. Single SDOF liner element using the LBM simulation setup and virtual probe locations for impedance calculations using Dean's method.

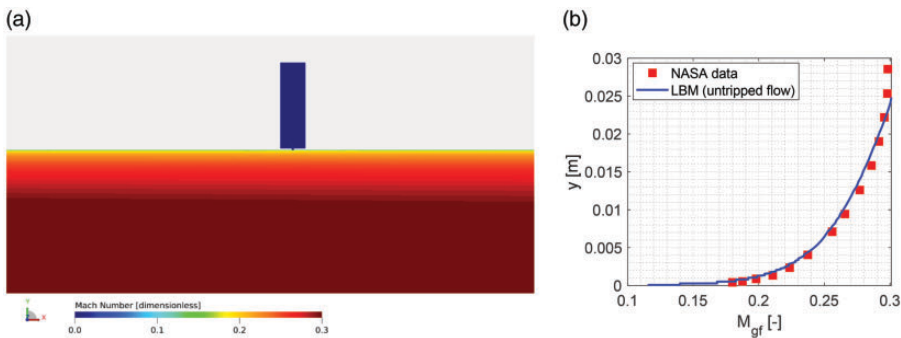


Figure 14. SDOF liner grazing flow simulation without boundary layer trip: (a) Instantaneous flow field and (b) time-averaged boundary layer profile upstream of the liner.

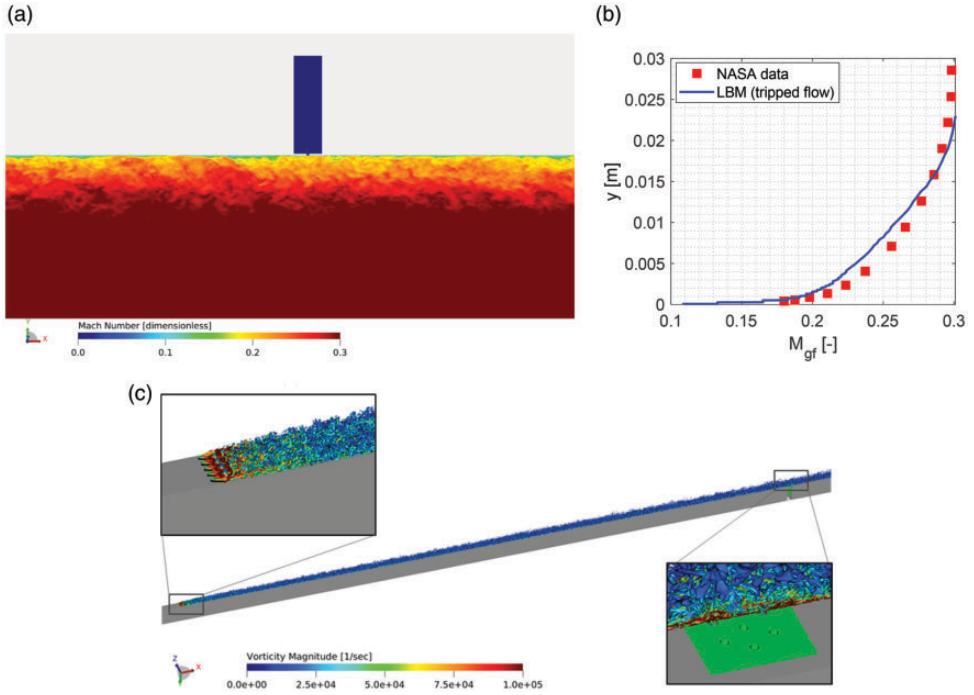


Figure 15. SDOF liner grazing flow simulation with boundary layer trip: (a) Instantaneous flow field, (b) time-averaged boundary layer profile upstream of the liner, (c) Snapshot of $\lambda_2 = -5e7 [1/s^2]$ iso-surface, colored by vorticity magnitude.

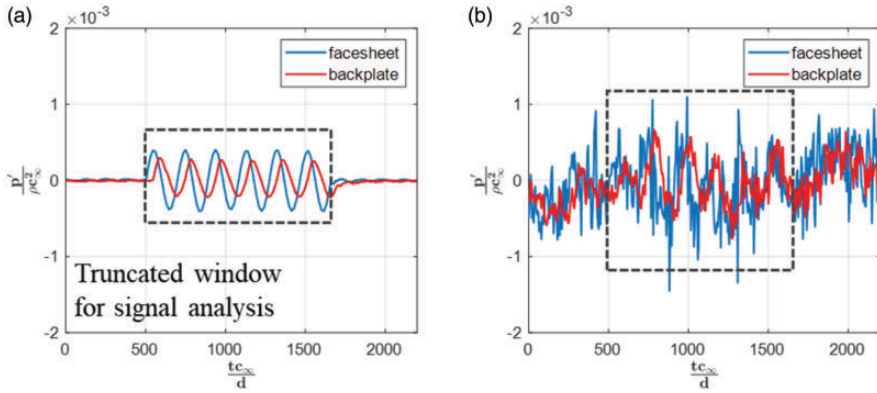


Figure 16. SDOF liner simulations time signal with $f = 1800$ Hz acoustic excitation in presence of grazing flow. (a) Untripped RANS-like flow and (b) Tripped VLES flow.

on the backplate of the cavity are investigated and shown in Figure 16 for an acoustic forcing at 1800 Hz. A clean sinusoidal signal associated with the single frequency acoustic forcing is visible in the untripped flow simulation for the duration of the time when the truncated wave (consisting of six cycles) passes over the liner (Figure 16(a)). In comparison,

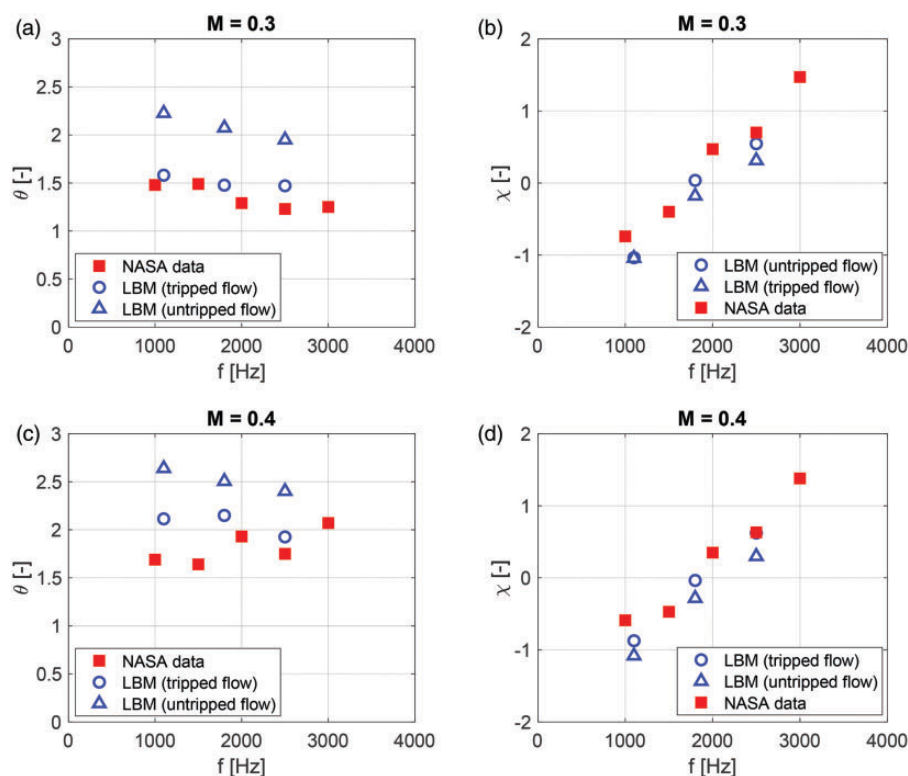


Figure 17. Grazing flow SDOF liner impedance predictions from high-fidelity LBM simulations for $\text{SPL} = 130$ dB, compared to NASA experimental data.

the tripped flow simulation leads to highly unsteady scale-resolving behavior with strong hydrodynamic fluctuations on top of the clean acoustic signal (Figure 16(b)). The acoustic signal is much more difficult to discern visually in the time signal in this case, but it is still clearly dominant in the frequency spectrum (not shown here). For both simulation approaches the same truncated time period of the recorded pressure signal is analyzed and the calculated impedance, via Dean's method, is averaged over all seven orifices. The results from this impedance calculation are shown in Figure 17, for grazing flow speeds of $M_{\text{gf}} = 0.3$ and $M_{\text{gf}} = 0.4$ and compared to NASA experimental data. For all cases, the predicted reactance is in good agreement with the experimental data; although, the untripped flow cases tend to consistently predict slightly lower reactance. Comparing resistance values, the tripped dynamic flow solution cases still agree fairly well with the NASA data, while the untripped RANS-like flow solution cases show a clear overprediction of the data. Based on this observation, it is concluded that a scale-resolving simulation is needed to predict the grazing flow effect correctly. It should be pointed out that the impedance in the NASA experimental data sets has been determined using an eduction method which involves an iterative process. Some of the differences observed between the LBM predictions and the experiment data may therefore be related to the different approaches used in computing the liner impedance.

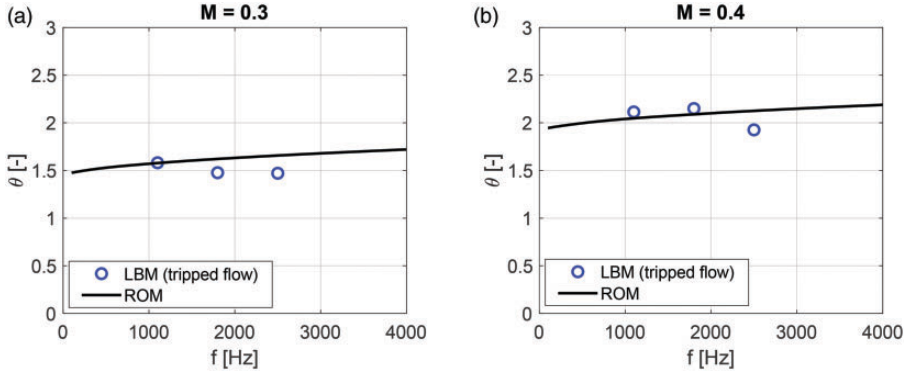


Figure 18. Grazing flow SDOF liner impedance predictions from high-fidelity LBM simulations for 130 dB excitation, compared to semi-empirical ROM (θ of equations (3) and (7)).

The high-fidelity LBM simulation results are now compared to the Goodrich model grazing flow impedance term, which is given by

$$z_{gf} = \frac{1}{\sigma} \frac{1}{2 + 1.256\delta^*/d} M_{gf} + i \left\{ \frac{\omega t}{c\sigma} \left[1 + \left(9 + \frac{(k_s a)^2}{2} \right)^{-0.5} + 2 \frac{8a}{3\pi t} \frac{1 - 0.7\sqrt{\sigma}}{1 + 305M_{gf}^3} \right] - \cot(kD) \right\} \quad (7)$$

where δ^* is the boundary layer displacement thickness. For a ratio of $\delta^*/d \sim 1$, the resistance term reduces to the well-known simpler form of $\theta_{gf} = \sigma^{-1} 0.3 M_{gf}$ (see Motsinger and Kraft³¹). This simplified version is used here and co-plotted with the high-fidelity LBM data, shown in Figure 18. The results are found to agree reasonably well.

As the source amplitude is increased, linear grazing flow effects and non-linear SPL effects start to combine, potentially competing. The combination of these two effects was investigated for $M_{gf} = 0.3$, using the tripped flow solution of Figure 15, and the results are shown in Figure 19. The ROMs for grazing flow effects (equation (7)) and non-linear SPL effects (equation (4)), when plotted together, predict a cross-over point near ~ 156 dB. Below this threshold, grazing flow effects are predicted to dominate and above this threshold the non-linear SPL effects are predicted to dominate. The LBM data confirms this trend. For SPL = 130 dB, the resistance agrees well with the grazing flow ROM. For higher SPL, there is a slight increase in resistance, with a more clearly noticeable larger rise beyond 150 dB and towards 160 dB. The effects of grazing flow and SPL are not additive. For low SPL, the resistance is determined by the grazing flow effects, whereas above a certain threshold, the SPL effects dominate. For $M_{gf} = 0.3$ this threshold is found to be rather high (> 150 dB). It is worth noting that a similar study was performed with the untripped flow field of Figure 14. The results are not shown here for brevity. It was found that the LBM results agree well only at 160 dB where the non-linear SPL effects dominate. This confirms the previous observation, that a RANS-type flow field overpredicts the grazing flow effect significantly, but once the SPL effects dominate the resistance, the predictions agree well with the tripped, scale-resolving flow simulations.

In summary, the leading order grazing flow impedance effects were investigated here and found to be predicted correctly using the scale-resolving LBM high-fidelity simulation

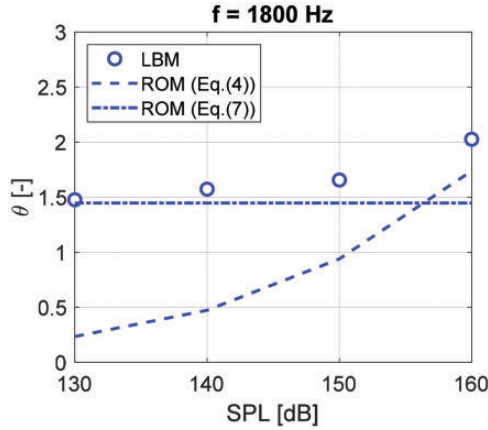


Figure 19. Grazing flow SDOF liner resistance predictions from high-fidelity LBM simulations for $M_{gf} = 0.3$ and for different SPL, compared to non-linear SPL ROM (equations (3) and (4)) and grazing flow ROM (equation (7)).

approach. A more detailed analysis of turbulent scale resolution requirements in the boundary layer of the oncoming grazing flow, as well as a detailed analysis of the orifice flow fields have not been conducted here, as they are outside the scope of this paper. The reader is referred to the on-going research by Avallone on this topic.³²

Bias flow

For most traditional nacelle liner applications, bias flow through the liner is not common and so typical industry-standard impedance models do not necessarily include terms for bias flow effects. Bias flow is most relevant in combustion applications where cooling flow through the liner facesheet is required or needed based on material constraints. However, the addition of bias flow has also been studied as a means to control the impedance in nacelle liners during different flight conditions^{33,34} and is therefore relevant to advanced nacelle liner concepts. To study the effect of bias flow through the facesheet of the SDOF liner using the high-fidelity LBM approach, the same grazing flow facility numerical setup (Figure 12) is used. The domain inlet velocity is set to zero, while a velocity boundary condition is applied at the backplate of the cavity. The simulations are again compared to an existing ROM. For bias flow, the widely known model of Howe²⁰ is used, with the addition of the reactance term $-icot(kD)$ to account for the presence of the backing cavity:

$$z_{bf} = \frac{1}{\sigma} \frac{\pi k a}{2(\gamma^2 + \delta^2)} (\delta + i\gamma) - icot(kD) \quad (8)$$

with

$$\gamma = \frac{I_1^2(St) \left(1 + \frac{1}{St}\right) + \frac{4}{\pi^2} \exp(2St) \cosh(St) K_1^2(St) \left(\cosh(St) - \frac{\sinh(St)}{St}\right)}{I_1^2(St) + \frac{4}{\pi^2} \exp(2St) \cosh^2(St) K_1^2(St)} \quad (9)$$

and

$$\delta = \frac{\frac{2}{\pi St} I_1(St) K_1(St) \exp(2St)}{I_1^2(St) + \frac{4}{\pi^2} \exp(2St) \cosh^2(St) K_1^2(St)} \quad (10)$$

I_1 and K_1 are modified Bessel functions, and $St = ka/M_{bf}$ is the orifice Strouhal number. Acoustic waves are still injected into the domain at grazing angle to the liner. The following study is focused on an excitation at a fixed frequency of 1800 Hz with the bias flow Mach number M_{bf} varied. For the excitation to be away from resonance, the cavity depth has been shortened by 50%. The steady flow field from the flow through the facesheet orifices is visualized by iso-surfaces of total pressure and shown in Figure 20 for two different bias flow Mach numbers. The bias flow jets appear largely steady near the orifice exit planes.

The predicted impedance values for a single tone excitation of 130 dB for different bias flow Mach numbers are plotted in Figure 21. The resistance increases strongly and linearly

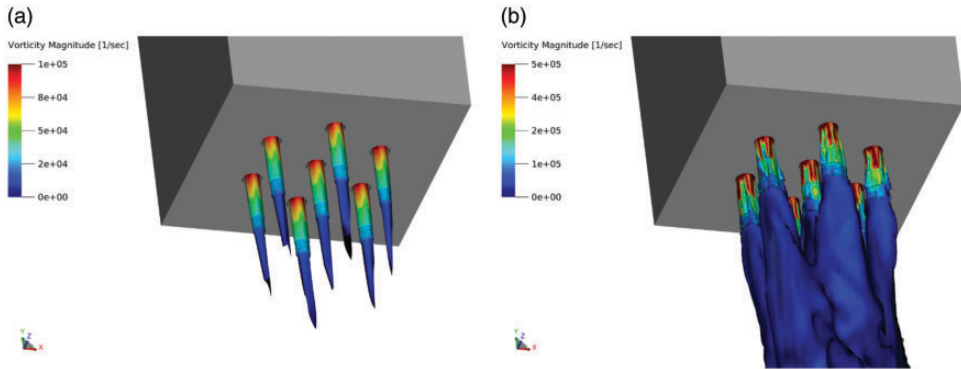


Figure 20. Total pressure iso-surfaces of 101,400 Pa colored by vorticity magnitude from LBM simulations for (a) $M_{bf} = 0.03$ and (b) $M_{bf} = 0.12$.

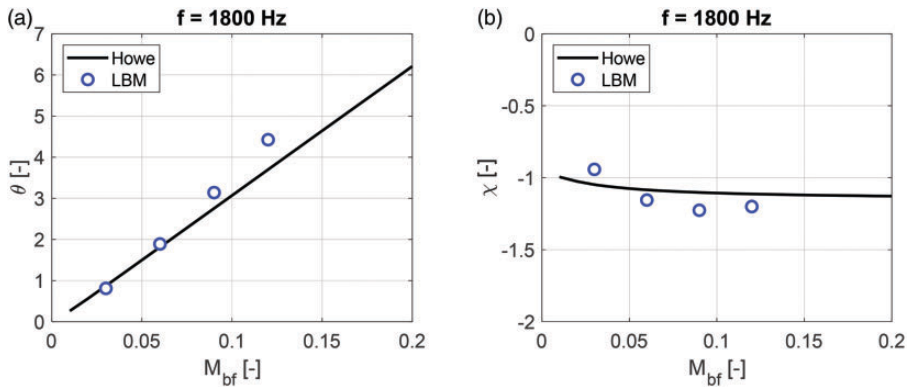


Figure 21. Bias flow SDOF liner impedance predictions from high-fidelity LBM simulations using a truncated sine wave excitation at 130 dB, compared to the analytical model by Howe (equation (8)).

with M_{bf} , whereas the effect on reactance is rather small. The high-fidelity simulation results are found to agree well with the analytical model of Howe for moderate values of M_{bf} .

At higher levels of acoustic excitation, the unsteady high acoustic particle velocity through the orifice combines with the steady mean flow and the resulting net effect is a combination of linear bias flow and non-linear orifice jet flow effects.^{35,36} To investigate this effect, the LBM simulations have been repeated at 1800 Hz for SPL of 140 dB, 150 dB, and 160 dB. The results for the predicted resistance are shown in Figure 22(a), whereas Figure 22(b) shows the corresponding predictions using the non-linear impedance model given by equations (3) and (4) and the linear bias flow model given by equation (8). It can be observed that, overall, the liner resistance appears to be a non-additive combination of SPL and bias flow effects. At low M_{bf} there is a region where the SPL effect dominates with an almost flat response, up to a cross-over point above which the bias flow effect dominates. For example, for $M_{bf}=0.06$ all of the LBM simulations predict close to the same resistance value—an indicator that the solution is independent of SPL and the bias flow effect dominates. The ROM plots in Figure 22(b) show that for $M_{bf}=0$ all predictions with the non-linear model given by equations (3) and (4) agree very well with the corresponding LBM predictions. As M_{bf} is increased, the LBM predictions show a slight increase in resistance, indicating the beginning of a blending or transition region that exists up to the point where the bias flow effect dominates the resistance behavior. This transition region appears to be more pronounced in the case of 160 dB excitation, as evidenced by the lower resistance values for $M_{bf} > 0.07$. It is expected that the 160 dB curve merges with the lower SPL curves when M_{bf} is sufficiently high (beyond the range studied here).

Grazing flow and bias flow

In all engine acoustic liner applications grazing flow is present and if bias flow is considered, it should not be treated in isolation. Indeed, even while some semi-empirical models assume the flow effects to be additive, it has been observed that this is not the case and grazing and bias flow interactions need to be taken into account.^{37,38} The picture becomes even more complicated when SPL effects are included in the combination of these effects.

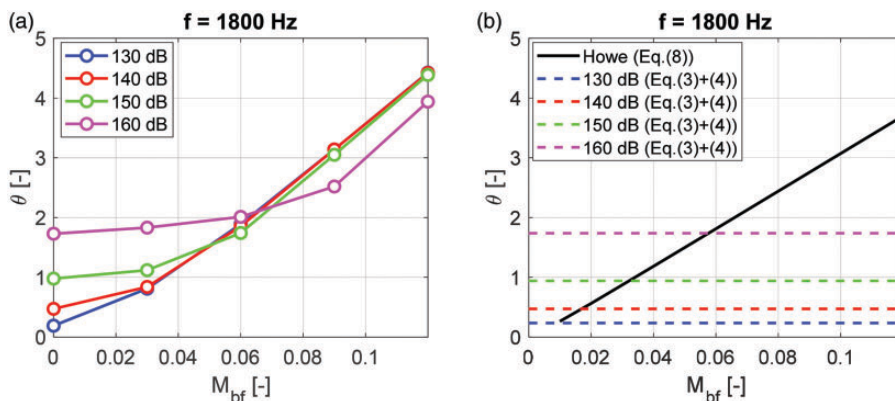


Figure 22. Bias flow SDOF liner impedance predictions from (a) high-fidelity LBM simulations for different excitation amplitudes and (b) ROM predictions.

For simplicity, only single frequency excitations at 1800 Hz with SPL = 130 dB are considered here. The combined flow effect is studied in the same numerical setup as was used previously to study grazing and bias flow effects in isolation. Simulations were performed for a fixed $M_{gf} = 0.3$ and for a range of M_{bf} . Snapshots of the flow fields around one of the seven facesheet orifices are shown in Figure 23 for different M_{bf}/M_{gf} ratios.

The acoustic impedance is calculated as previously, using equation (6), and results are shown in Figure 24. For small values of M_{bf}/M_{gf} the resistance is dominated by grazing flow effects and is insensitive to the amount of bias flow. The resistance in this range is predicted well by the ROM given by equation (7). As the bias flow is increased, the resistance increases and starts to follow the slope of the bias flow curve predicted using Howe's model given by equation (8). Similar to the SPL and bias flow combination, there is no pure additive effect of grazing and bias flow. Depending on the ratio, one has a more dominant effect on resistance than the other and there appears to be a small region of transition. Note that the bias flow prediction model of Howe has been shifted by $M_{bf}/M_{gf} = 0.28$ on the abscissa of Figure 24(a). Similar values on the order of ~ 0.3 for the demarcation between grazing and bias flow effect dominance have been reported in the literature.³⁷ The comparison between LBM and the two ROMs for the reactance is shown in Figure 24(b). Neither of the ROMs predict the observed change in reactance with increased bias flow.

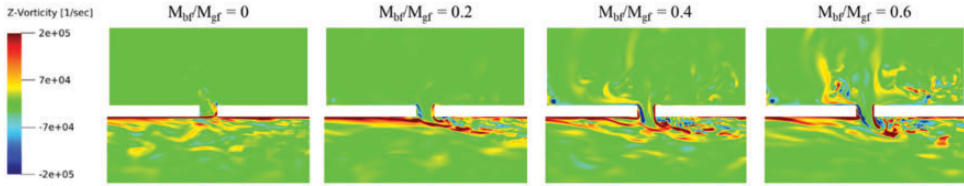


Figure 23. Instantaneous vorticity contours of the flow field around one of the seven liner facesheet orifices shown for different grazing-to-bias flow Mach number ratios.

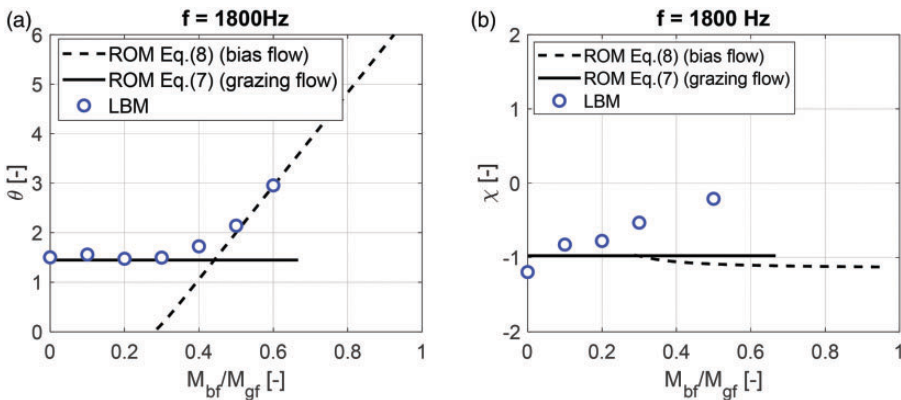


Figure 24. Combined bias and grazing flow SDOF liner impedance predictions from high-fidelity LBM simulations for single frequency excitation at 130 dB, compared to ROM: (a) resistance and (b) reactance.

Summary and conclusions

It was demonstrated through various test cases, including the effects of high SPL, grazing and bias flow, and combinations thereof, that high-fidelity LBM simulations are capable of predicting the various physical effects on impedance correctly. The focus was on a comparison of established ROMs, like the Goodrich model, to show the consistency between the empirically based tools and modern high-fidelity simulation methods for a traditional SDOF liner with sharp-edged orifices. When combinations of different effects occur (like bias and grazing flow, for instance), it was found that the effects are not additive and so simply adding up standard ROMs for resistance and reactance is not appropriate. This provides an opportunity for high-fidelity LBM simulations to improve existing ROMs to include interaction effects. Such interaction effects are expected to become more important for advanced acoustic liner concepts with complex liner geometries. On the one hand, such high-fidelity approaches are capable of providing further insight into the physical mechanisms responsible for the different behaviors observed and are well-suited for studies of advanced and more complicated liner topologies. On the other hand, for industrial practice a more rapid “mid-fidelity” liner design screening approach is needed. The main idea is that existing semi-empirical models can be embedded in such a mid-fidelity approaches and can be further augmented and updated by high-fidelity simulations, as needed. These two simulation approaches are therefore meant to supplement each other when studying advanced liner concepts. The mid-fidelity tools can be exercised in a first step to assess advanced liner concepts and design variants to capture trends and help guide the design process. Once an initial design is converged to, the high-fidelity approaches would be used as a check on the overall performance of the design and would be considered the answer closer to the truth, in the absence of experimental data. The applicability of mid-fidelity tools to capture more complex interaction effects, like combinations of high SPL, grazing, and bias flow, still needs to be established and currently limit their use for particular design requirements until improved sub-models, that account for these effects, can be integrated.

Advanced liner concept evaluation

Overview

The design of an advanced liner concept is described here to highlight the use of the previously described high-fidelity LBM simulation approach towards more complex liner concepts, and how one might be able to use a more sophisticated ROM to predict leading order effects via a mid-fidelity simulation approach. The analysis presented here is focused on the NASA NIT setup to showcase the application. The analysis can be easily extended towards more complex simulations with flow and SPL effects in a manner similar to what was described in the previous section.

Description of the liner structure

In the beginning, there was a desire to explore surfaces that could be easily prescribed and parameterized. One such set of surfaces is the triply periodic minimal surfaces (TPMS) set. They are mathematically well-defined periodic and non-self-intersecting surfaces that result in partitioned volumes consisting of two intertwined congruent labyrinth structures. The surfaces themselves represent highly flexible, adaptable, and versatile building blocks for

generating more complex structures for material designs.³⁹ Some examples of TPMS structures are shown in Figure 25.

Recently, TPMS structures have also gained attention as acoustic metamaterials whose effective acoustic properties depend largely on the topology rather than composition.^{40,41} In the present study, the Schwarz P TPMS, shown in Figure 25, has been chosen as the base geometry for an advanced metamaterial acoustic liner design. Figure 26 shows an illustration of this concept. This advanced liner design consists of a Schwarz P TPMS structure integrated in the cavity with a perforated facesheet in front. The TPMS structure contains interconnected air volume cells with a large number of degrees of freedom for tuning the absorption characteristics. This approach, therefore, offers a convenient framework for realizing linear performance for lower frequency and more broadband liner attenuation.

Performance predictions

The acoustic absorption performance of a set of Schwarz P TPMS metamaterial liner designs was predicted using the high-fidelity LBM simulation approach, as described in the earlier section, and also using a mid-fidelity simulation approach, which may be viewed as a more sophisticated ROM. Due to its relative efficiency in setting up the model and its lower computational load, the mid-fidelity modeling approach, using the FE acoustic solver Actran, is

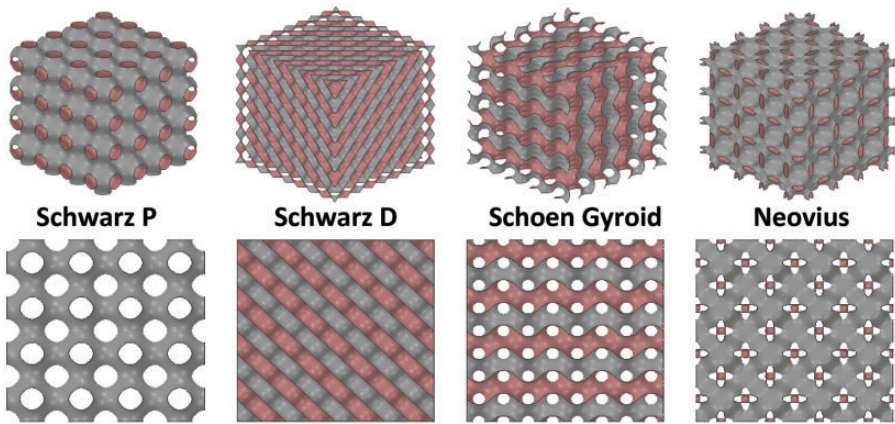


Figure 25. Examples of TPMS geometries (iso and front view).

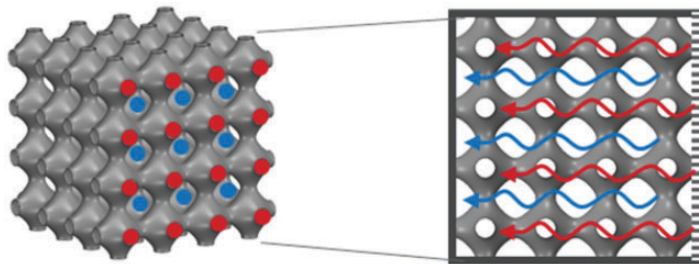


Figure 26. An acoustic liner design incorporating a Schwarz P TPMS structure.

suitable for more rapid screening and trial of various TPMS liner designs. In the implementation of this approach in Actran, the perforation models were applied in the form of transfer admittances, which relate the particle velocity and the differential acoustic pressure across the perforation. This approach is therefore similar to the approach used by Schiller et al.⁷ and Sugimoto et al.¹⁰ as described in the literature review section. The transfer admittance (or impedance) is calculated offline and thus can be based on any semi-empirical perforation model. In the present study, the linear perforation model of equation (3) was used without the cavity reactance term $-icot(kD)$. Using the mid-fidelity modeling approach first, multiple design iterations were performed, but only two designs will be shown here. The normal incident absorption performance for the two preliminary TPMS liner designs, for the no flow condition ($M = 0$) and for a broadband excitation with OASPL = 130 dB, as predicted by the mid-fidelity modeling approach and the high-fidelity LBM approach, are shown in Figure 27. The absorption coefficient α was determined from the predicted impedance as follows:

$$\alpha = \frac{4\theta}{(1 + \theta)^2 + \chi^2} \quad (11)$$

In Figure 27, the design shown in blue color consisted of four bulb layers with a total liner depth of 1.5 in, while the other design shown in green consisted of two bulb layers with a total liner depth of 0.75 in. The liner depth of 1.5 in for the four-bulb design is consistent with the depth of the more traditional reference SDOF liner shown in Figure 7(a), for which prediction results are co-plotted in Figure 27 (dashed black curve). As shown by the prediction results from both the mid- and high-fidelity simulations in Figure 27, the TPMS metamaterial concept of the same material depth (i.e., same volume) shows peak absorption at a much lower frequency (blue curves) than that of the reference SDOF liner of the same depth (black dashed curve). This frequency can be shifted to match the SDOF liner by reducing the number of bulb layers which is represented by the adjusted concept consisting of two bulb layers and which shows superior performance to the SDOF (depth of 1.5 in) with the same peak absorption frequency (green curves). The desired, higher compactness of the latter design for the same peak frequency translates into a volume reduction of $\sim 50\%$. The results highlight that the TPMS metamaterial concept is able to use the available

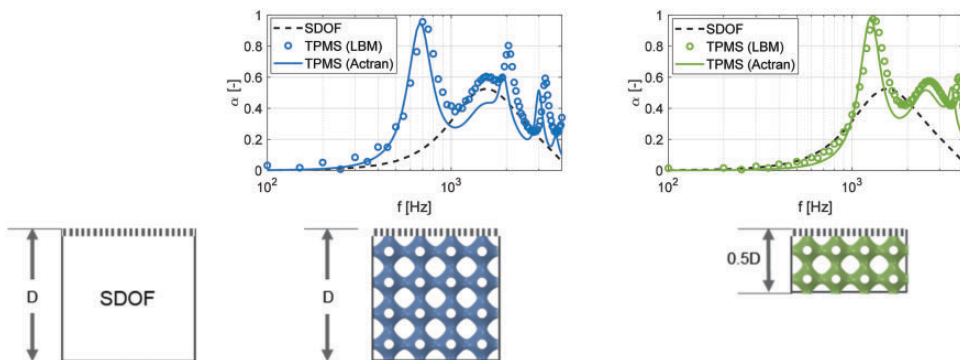


Figure 27. TPMS metamaterial absorption compared to traditional SDOF liner, targeting 1.5 kHz for $M = 0$.

volume more efficiently than a traditional SDOF liner, thus demonstrating its potential. In addition, this concept provides higher peak absorption values and increased broadband absorption for frequencies higher than the peak absorption frequency. In terms of the prediction results, both mid- and high-fidelity simulations predict very similar performance for both TPMS concepts at this OASPL, for which the liner performance is expected to be linear (see Figure 9). It is apparent, though, that the high-fidelity LBM simulations show a slight shift of the absorption curve in frequency towards higher values. The reason for this difference might be related to the finite wall thickness used in the LBM simulations, whereas the mid-fidelity Actran simulations were based on TPMS surfaces of zero thickness. The observed discrepancies between LBM and Actran at higher frequencies need to be further investigated through validation with test data. This is where the benefits of the high-fidelity LBM simulations are apparent: the actual geometry can easily be represented and can include the fine features such as the orifice and edge shapes (sharp, rounded, etc.) which influence the physics involving complex interactions between flow and acoustics in and around such features. In contrast, a mid-fidelity tool like Actran has to depend on semi-empirical models for this, and the adequacy for some of these models for complex geometries and interaction physics still needs to be established. The actual liner geometry, which may now be produced by additively manufactured processes, could be obtained from a computerized tomography (CT) scan and thus allowing the high-fidelity LBM approach to be used to include manufacturing-related geometry changes and assess the sensitivity to acoustic performance.⁴² The authors plan to continue to design and fabricate these structures and intend to follow this publication with additional test results and findings.

Conclusions

Aircraft engine acoustic liners of the future are expected to consist of more complex structures to achieve the lower and broader frequency attenuation performance and do so with a smaller (more compact) liner footprint than the traditional liner constructions of today. Innovative liner concepts will need new and validated methods for design, construction, and analysis to ensure a viable path for optimization and ultimately certification in engines and aircraft of the future. As indicated, the goals of this paper were threefold: (1) document the use and validity of current approaches along with identifying the fundamental physics, (2) in a systematic way, establish tools/approaches for efficient and effective designs and evaluations of future novel and more complex acoustic liners, and (3) demonstrate, with a simplified example, the design process for one such novel liner concept. It was demonstrated that a high-fidelity LBM VLES approach is capable of predicting all major absorption mechanisms of a SDOF liner correctly, for low and high SPLs, grazing and bias flow, and combinations thereof. The application of this simulation approach to a novel advanced liner concept was made to illustrate its use. The approach lends itself to assessing manufacturing effects through directly importing manufactured and CT scanned test coupons and to build more sophisticated ROMs for liner impedance relevant to advanced concepts.

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Appendix

Notation

a	perforate hole radius
c	speed of sound
d	perforate hole diameter
C_d	discharge coefficient
D	liner cavity depth
k	acoustic wavenumber
k_s	shear wavenumber
M_{gf}	grazing flow Mach number
M_{bf}	bias flow Mach number
t	facesheet thickness
R	reflection coefficient
St	Strouhal number
z	normalized acoustic impedance
α	absorption coefficient
ρ	density
θ	normalized resistance
χ	normalized reactance
DDOF	double degree of freedom
LBM	Lattice Boltzmann Method
OASPL	overall sound pressure level
POA, σ	percent open area
ROM	reduced order model
SDOF	single degree of freedom
SPL	sound pressure level
VLES	very large eddy simulation